

VIMStm Utilization & Operator Training

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

VIMStm Utilization & Operator Training.....0

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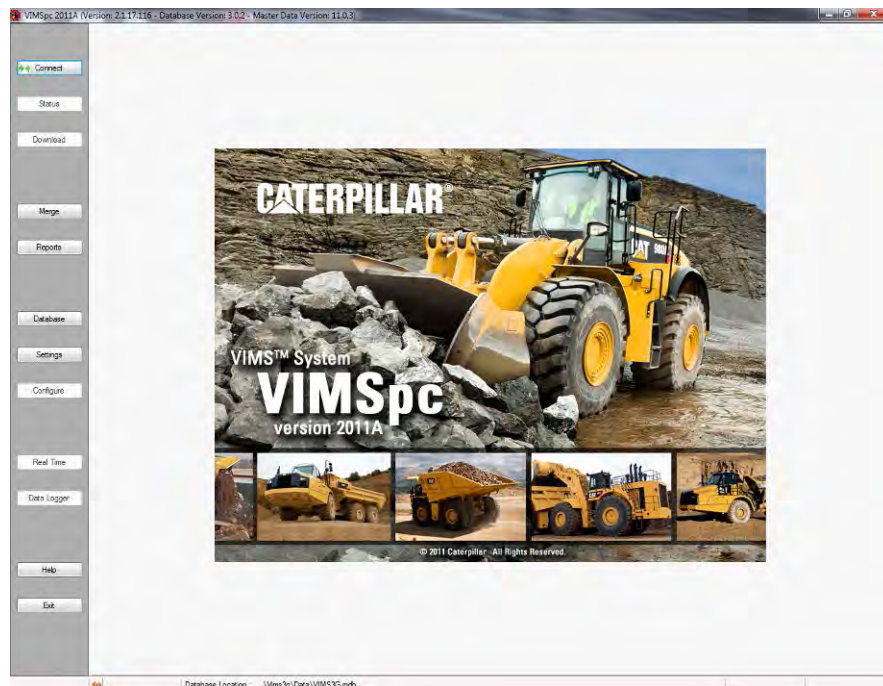
1.0 Introduction

Operator training is critically important to a mine as it can have a real, tangible impact on a site's overall cost per ton and profitability. Operating Cat equipment within its design parameters will have a positive impact on the physical availability and productivity metrics for the mine. However, despite the importance and impact of this topic, much variation is seen from mine to mine when considering the amount of feedback and instruction that takes place after the initial training course is provided or after a new machine is delivered to the site. This best practice focuses on a tool that may be utilized to a greater degree by many mines to further enhance the operator training program, which is the utilization of VIMS[™] data.

2.0 Best Practice Description

Mining customers may regularly download and analyze VIMS[™] data in the Maintenance department to determine how machine components and systems are currently functioning in order to proactively preempt an unplanned repair. Or, the Maintenance department may download and analyze data to determine how components and/or systems were behaving before a failure occurred as part of the failure analysis process. Additionally, the Operations department may analyze VIMS[™] data to understand productivity measurements such as total number of cycles, payload hauled per cycle, cycle time, average loader passes, estimated fuel rate per cycle, etc. Although these are good, legitimate uses of VIMS[™] data, the Training department is another area of mine management that can benefit from the effective use of VIMS[™] data.

If the mine management system MineStar is used, VIMS[™] data is wirelessly and continuously streamed to the mine controller's office. If the mine does not have the capability to wirelessly stream data back to the office, VIMS[™] information can be downloaded manually by a service technician. If data is captured in this fashion, a laptop will be needed which has VIMSpc or VIMS 2011A installed, along with a VIMS[™] cable.



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Once the data arrives at the office, it can then be accessed by the Training department supervisor for analysis. Some key areas to consider for analysis include the following:

- Event List: this is a record of all the events / diagnostics that have occurred on a machine, retaining the last 500 machine and system events. Some of the events which can be operator-induced may include:
 - o Park Brake
 - o Brake Stroke
 - o Payload Overload
 - o Payload Overload Abuse
 - o Payload Overload Dump
 - o Engine Speed
 - o Abusive Shift
 - o Gear Select
 - o Coast in Neutral
 - o Body Position
 - o Body Up / Ground Speed
 - o Fuel Level

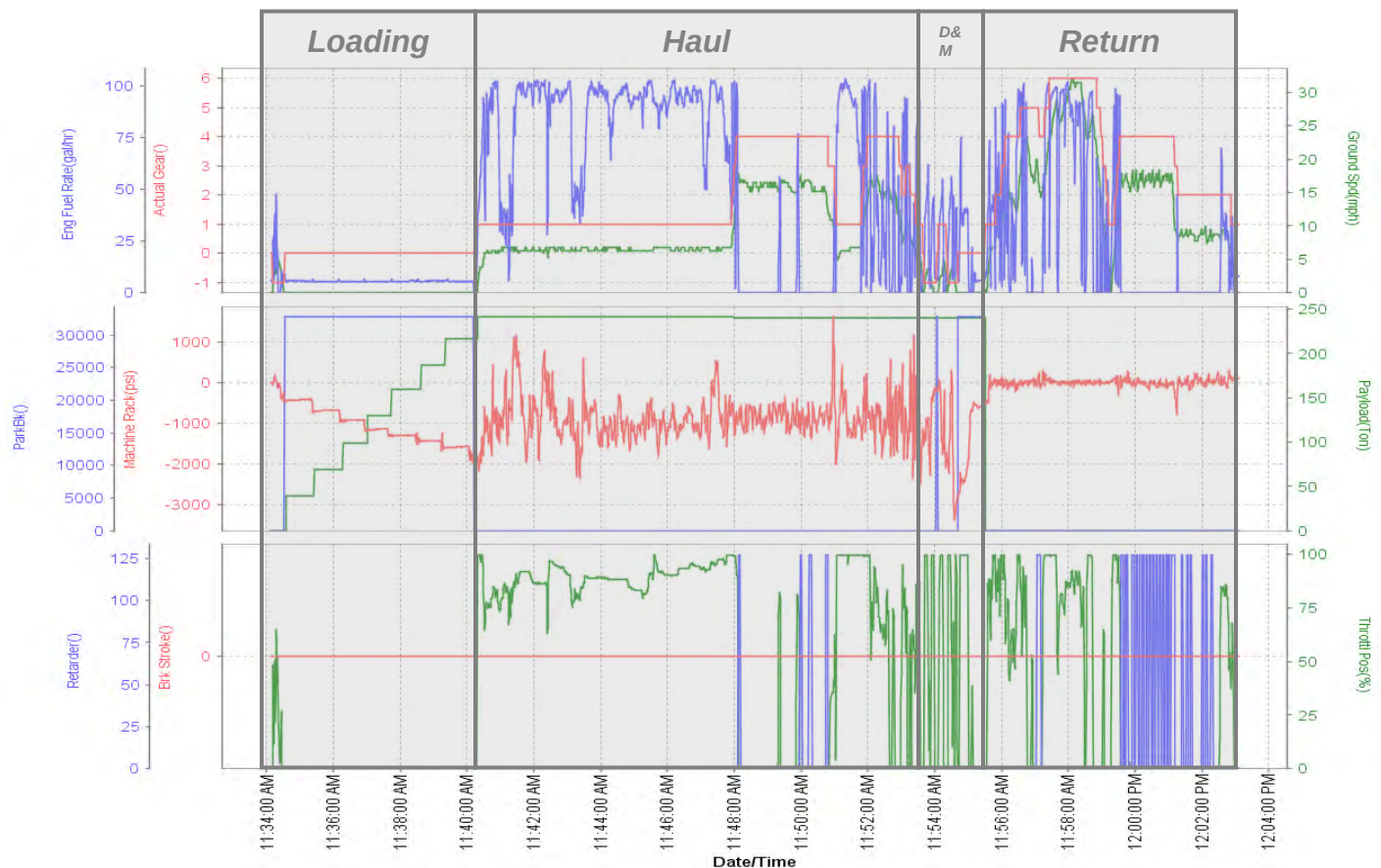
Time/Date	SMH	Event Id	Description(ID)	Limit	Worst Value MID-CID-FMI	Event Level	Duration	Ack #	Ack Time	DL/SS Link
06/26/2012 11:29:19 AM	41866	2915	Pyld Ovrld	0 Ton	288 Ton	2	00:09:10	1	400	
06/26/2012 11:48:43 AM	41867	2915	Pyld Ovrld	0 Ton	283.5 Ton	2	00:10:00	0	0	
06/26/2012 12:50:02 PM	41867	2915	Pyld Ovrld	0 Ton	278.1 Ton	2	00:00:49	0	0	
06/26/2012 02:39:25 PM	41868	2915	Pyld Ovrld	0 Ton	287.5 Ton	2	00:09:27	0	0	
06/26/2012 04:42:55 PM	41870	2915	Pyld Ovrld	0 Ton	286.2 Ton	2	00:00:54	0	0	
06/26/2012 04:57:01 PM	41870	76	Gear Select		Neutral	2	00:00:04	0	0	
06/26/2012 09:00:45 PM	41874	2915	Pyld Ovrld	0 Ton	283.7 Ton	2	00:09:24	1	23	
06/26/2012 10:06:11 PM	41875	2915	Pyld Ovrld	0 Ton	275.7 Ton	2	00:01:10	1	47	
06/27/2012 09:43:33 AM	41879	2915	Pyld Ovrld	0 Ton	277.3 Ton	2	00:00:16	1	6	
06/27/2012 10:08:35 AM	41879	2915	Pyld Ovrld	0 Ton	275.6 Ton	2	00:00:15	1	5	
06/27/2012 10:36:41 AM	41880	2915	Pyld Ovrld	0 Ton	274.7 Ton	2	00:00:13	1	6	
06/27/2012 07:25:12 PM	41887	2915	Pyld Ovrld	0 Ton	288.9 Ton	2	00:07:05	1	10	
06/27/2012 07:45:24 PM	41888	2915	Pyld Ovrld	0 Ton	275.4 Ton	2	00:04:50	1	6	
06/28/2012 06:05:18 AM	41895	2915	Pyld Ovrld	0 Ton	280.2 Ton	2	00:02:12	1	19	
06/28/2012 11:54:11 AM	41900	2915	Pyld Ovrld	0 Ton	276.9 Ton	2	00:00:18	1	14	
06/28/2012 11:43:30 PM	41909	2915	Pyld Ovrld	0 Ton	277.3 Ton	2	00:00:14	1	5	
06/29/2012 04:23:15 AM	41912	2916	Pyld Ovrld Dump	0 Ton	314.5 Ton	2	00:00:26	1	18	
06/29/2012 04:23:42 AM	41912	2917	Pyld Ovrld Abuse	0 Ton	314.5 Ton	2	00:00:04	0	0	
06/29/2012 04:31:36 PM	41921	2915	Pyld Ovrld	0 Ton	292.8 Ton	2	00:00:14	0	0	
06/29/2012 09:08:03 PM	41925	2915	Pyld Ovrld	0 Ton	279.7 Ton	2	00:00:13	1	7	
06/30/2012 12:46:49 AM	41929	2915	Pyld Ovrld	0 Ton	279.2 Ton	2	00:13:38	2	5	
06/30/2012 05:07:52 AM	41933	601	Payload Data	5 %	0 %	1	47:54:24	14	13	
06/30/2012 11:30:19 AM	41939	2915	Pyld Ovrld	0 Ton	279.8 Ton	2	00:13:44	2	12	
06/30/2012 09:19:44 PM	41947	83	Fuel Lvl	10 %	7.59 %	2	00:04:20	1	153	
06/30/2012 11:46:28 PM	41949	2915	Pyld Ovrld	0 Ton	296.7 Ton	2	00:09:36	1	5	
07/01/2012 12:01:57 AM	41950	2915	Pyld Ovrld	0 Ton	274.6 Ton	2	00:01:18	1	5	
07/01/2012 12:39:17 AM	41950	2915	Pyld Ovrld	0 Ton	279.4 Ton	2	00:09:08	1	4	
07/01/2012 07:19:02 AM	41956	2915	Pyld Ovrld	0 Ton	277.2 Ton	2	00:07:59	1	7	
07/01/2012 02:34:21 PM	41962	711	Lt R Brk Temp	255.2 °F	258.8 °F	2	00:00:10	0	0	
07/02/2012 06:35:38 AM	41977	2915	Pyld Ovrld	0 Ton	286 Ton	2	00:00:33	1	5	

Image of an Events List detailed report indicating payload overloads and abuse, gear select (neutral), and low fuel level.

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- **Data Logger:** captures real-time machine data similar to the event recorder, but must be activated by the operator or service technician. When taking a Data Logger, data is collected from all VIMS™ parameters at a sampling rate of once per second, and a maximum of 30 minutes of data can be stored. Operator trainers and shift supervisors can utilize this tool to help understand how an operator's technique may be impacting the machine. When using this feature, it is recommended to take several Data Loggers of the same scenario (i.e. truck cycles from loading tool to the dump point, and the return) to ensure the data captured represents reality. Some useful parameters to graph from a Data Logger include:

- o Actual Gear
- o Brake Stroke
- o Engine Fuel Rate
- o Engine Speed
- o Ground Speed
- o Machine Rack
- o Max Bias / Second
- o Max Rack / Second
- o Park Brake
- o Payload
- o Retarder
- o Throttle Position



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Image of a Data Logger graph example indicating throttle variation and machine rack spikes in load & dump areas, and 1st gear on approximately half the haul road.

- Cumulative: show number of occurrences of specific events over the life of the machine. Some useful occurrences to track which can be operator-induced are:
 - o Number of Abusive Shifts – 1st to Reverse
 - o Number of Abusive Shifts – Neutral to 1st
 - o Number of Abusive Shifts – Neutral to Reverse
 - o Number of Abusive Shifts – Reverse to 1st
 - o Engine Time @ Idle
 - o Estimated Fuel Used Idling
 - o Estimated Total Fuel Used
 - o Time in Gear (depends on machine, but all gears should be analyzed including neutral and reverse)

Cumulative Detail – Serial Number: FDBnnnnn

Description	Data Value
# of Abshift-1 to R	4.0 count
# of Abshift-N to 1	26.0 count
# of Abshift-N to R	9.0 count
# of Abshift-R to 1	145.0 count
Eng Time @ Idle	2622.38 hour
Est Fuel Used Idling	6.91 gal
Est Total Fuel Used	1183.47 gal
Time In 1C Gear	307.52 hour 4.7%
Time In 1L Gear	485.6 hour 7.3%
Time In 2nd Gear	748.38 hour 11.3%
Time In 3rd Gear	420.74 hour 6.4%
Time In 4th Gear	648.74 hour 9.8%
Time In 5th Gear	617.35 hour 9.3%
Time In 6th Gear	501.57 hour 7.6%
Time In Neutral	2743.46 hour 41.5%
Time In Reverse	138.83 hour 2.1%

Image of a Cumulative report indicating abusive shift occurrences and time spent in different gears. Note: for a particular site, 41.5% idle time may be acceptable / reasonable; further analysis would need to be done given the application.

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- Truck Payload: utilizes strut pressure technology for payload measurement. The truck system stores up to 2,400 payload cycles, cycle times, distance, time, and date of each payload. Some of the factors that may be beneficial to track in order to determine deviations from the average are:
 - o Payload
 - o Load Time
 - o Stopped Empty Time
 - o Stopped Loaded Time
 - o Cycle Time
 - o Fuel Rate (estimate)

Time/Date	Payload SMH	Serial Number	Payload (Tons)	Travel Empty Time (d: hh:mm:ss)	Travel Empty Distance (Mi)	Stopped Empty Time (d: hh:mm:ss)	Load Time (d: hh:mm:ss)	Stopped Loaded Time (d: hh:mm:ss)	Loaded Travel Time (d: hh:mm:ss)	Loaded Travel Distance (Mi)	Cycle Time (d: hh:mm:ss)	Cycle Distance (Mi)	Loader Passes	Fuel Rate (Gal/Hr)
06/25/2011 03:09:18 PM	19143		274.642	00:03:03	0.4	00:12:08	00:02:58	00:01:48	00:20:16	2.6	00:40:13	3	6	56.693
06/25/2011 03:49:34 PM	19144		262.514	00:06:27	2.3	00:00:14	00:03:33	00:00:50	00:25:12	4.1	00:36:16	6.4	5	70.313
06/25/2011 04:25:50 PM	19144		318.192	00:09:35	3.4	00:00:23	00:02:59	00:01:02	00:14:57	1.8	00:28:56	5.2	5	56.509
06/25/2011 04:54:45 PM	19145		292.282	00:04:24	1.4	00:00:14	00:03:45	00:00:53	00:27:10	4	00:36:26	5.4	6	72.461
06/25/2011 05:31:11 PM	19146		249.173	00:14:07	4.2	00:10:41	00:03:09	00:01:09	00:18:01	2.4	00:47:07	6.6	5	48.709
06/25/2011 06:18:17 PM	19146		287.982	00:08:10	2.3	00:01:48	00:04:14	00:01:14	00:18:21	2.4	00:33:47	4.7	6	60.385
06/25/2011 06:52:03 PM	19147		235.061	00:07:24	2.4	00:03:59	00:02:38	00:01:48	00:18:42	2.5	00:34:31	4.9	4	55.625
06/25/2011 07:26:34 PM	19148		300.11	00:06:08	2.4	00:00:07	00:04:01	00:07:40	00:19:30	2.5	00:37:26	4.9	5	60.508
06/25/2011 08:47:32 PM	19148		257.332	00:08:22	1.8	00:18:59	00:02:28	00:00:56	00:18:21	2.4	00:49:06	4.2	5	43.075
06/26/2011 04:10:30 AM	19151		219.515	00:03:05	0.2	00:09:37	00:01:25	00:01:17	00:13:41	1.7	00:29:05	1.9	3	83.553
06/26/2011 04:39:38 AM	19151		254.355	00:08:14	1.7	00:15:47	00:02:48	00:01:04	00:13:40	1.7	00:41:33	3.4	5	34.296
06/26/2011 05:21:10 AM	19152		266.042	00:07:56	1.7	00:01:39	00:02:20	00:01:08	00:14:26	1.7	00:27:29	3.4	5	49.666
06/26/2011 05:48:39 AM	19152		251.268	00:06:04	1.7	00:01:33	00:02:18	00:02:28	00:13:36	1.7	00:25:59	3.4	4	49.647
06/26/2011 06:14:38 AM	19153		247.409	00:06:10	1.7	00:02:00	00:02:11	00:01:20	00:13:26	1.7	00:25:07	3.4	5	51.36
06/26/2011 06:39:45 AM	19153		257.993	00:06:33	1.7	00:01:43	00:04:13	00:01:07	00:13:22	1.7	00:26:58	3.4	5	50.062
06/26/2011 07:06:41 AM	19154		255.458	00:05:37	1.7	00:00:18	00:04:14	00:01:35	00:13:48	1.7	00:25:32	3.4	4	51.697
06/26/2011 07:32:11 AM	19154		244.873	00:05:26	1.7	00:00:08	00:03:10	00:21:37	00:14:34	1.7	00:44:55	3.4	4	33.729
06/26/2011 08:36:54 AM	19155		267.806	00:05:06	1.3	00:03:56	00:03:03	00:04:58	00:10:35	1.4	00:27:38	2.7	6	41.255
06/26/2011 09:04:31 AM	19155		265.38	00:08:49	1.6	00:09:46	00:03:34	00:01:09	00:22:35	2.6	00:45:53	4.2	5	46.422
06/26/2011 09:50:23 AM	19156		250.937	00:08:27	2.4	00:02:06	00:02:31	00:10:07	00:11:57	1.4	00:35:08	3.8	4	35.009
06/26/2011 10:25:30 AM	19157		301.323	00:05:08	1.4	00:00:21	00:02:19	00:01:27	00:23:53	2.6	00:33:08	4	4	64.738
06/26/2011 10:58:37 AM	19157		243.881	00:09:33	2.4	00:06:06	00:02:32	00:01:06	00:09:51	1.2	00:29:08	3.6	5	36.041
06/26/2011 11:27:44 AM	19158		308.6	00:06:48	1.4	00:03:51	00:03:16	00:05:54	00:25:39	3	00:45:28	4.4	5	53.446
06/26/2011 01:32:27 PM	19158		248.181	00:10:46	2.6	00:28:25	00:01:54	00:18:16	00:22:31	2.7	01:21:52	5.3	3	31.698
06/26/2011 02:54:18 PM	19160		256.229	00:08:09	2.4	00:02:39	00:02:33	00:01:08	00:10:00	1.3	00:24:29	3.7	5	44.724
06/26/2011 03:18:45 PM	19160		296.031	00:06:33	1.7	00:03:17	00:02:41	00:03:29	00:14:30	1.7	00:30:30	3.4	6	49.672

Image of a Payload Detail report indicating payload, stopped empty, stopped loaded, and estimated fuel rate. When analyzing this data, one needs to ensure similar data such as cycle distance and travel times is compared.

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3.0 Implementation Steps

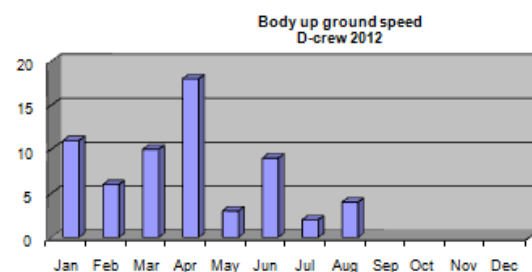
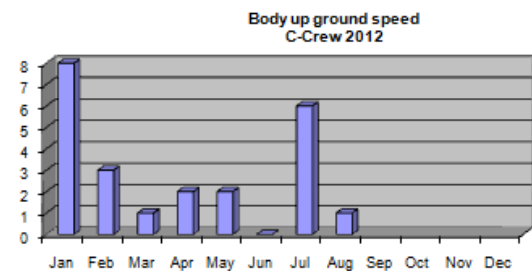
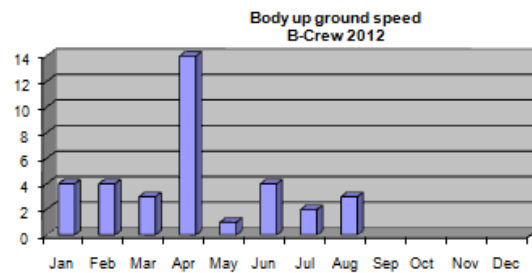
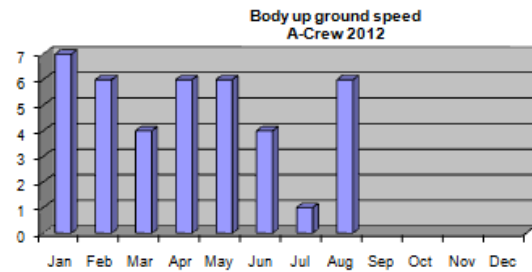
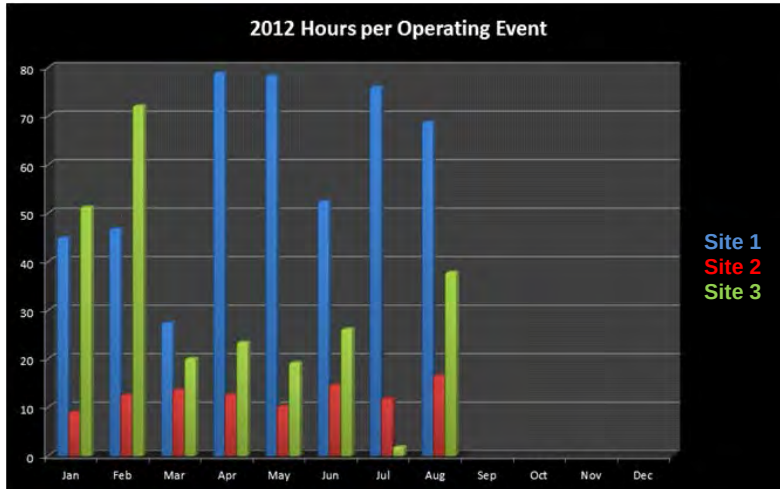
In order to be able to analyze the VIMS[™] reports mentioned above, the following steps need to be in place at a mine site:

1. Develop a regular download process of VIMS[™] data. This can be either a manual process (using the VIMS[™] cable mentioned below in Resources Required section), or performed wirelessly using a mine management system such as MineStar which can communicate the data back to the mine office. If the process is done manually, data should be downloaded on a weekly basis or more frequently to ensure analysis that is performed is current and relevant. To establish a wireless process at the site, additional information can be provided by the local Cat dealer.
2. Establish a destination folder on the shared drive where VIMS[™] data can be stored for the life of the machine. Access to this one-safe-source of VIMS[™] data can be restricted to those who will be using the data to produce information and reports.
3. Designate an individual (or individuals depending on the size of the fleet and corresponding workload) to perform the analysis of the VIMS[™] operational data. This individual should be familiar with the mine and equipment operation, as well as have an understanding of VIMS[™], its functionality, and its capability.
4. Determine what parameters will be analyzed and what metrics will be tracked. This determination should be made after seeking input from all user groups of the information, such as the Training, Operations, and Maintenance departments.
5. Analyze the parameters and produce the reports as agreed upon in the previous step. Analysis and reporting should be done at least once per week to show the importance of the initiative, keep the information in front of the intended audience, and trend results which reveal improvement or decline in the tracked metrics. Some sites may want to do this after every shift in order to analyze what has recently occurred on the last shift. These reports should be provided to the Shift Supervisor, Training Manager, Operations Manager, Maintenance Manager, and others who should be made aware.
6. Run comparison reports to determine how one crew compares to the others, and also possibly how one site compares to sister sites the company may have. This level of analysis, which can be produced on a weekly or monthly basis, can highlight training opportunities for particular crews and/or sites which are experiencing a relatively higher number of events. However, the inverse could also occur where good practices from one crew or site could be replicated to other crews or sites in order to lower the overall number of events, increase the overall productivity of the equipment, and lower the cost per ton.

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Depending on the level of technological automation at site, these reports can take from 30 minutes to one hour to produce. Some examples of these reports are found below:



	Level	Crew A	Crew B	Crew C	Crew D	Total
Abusive Shift alarm	Level 2	44	78	69	29	220
Body Up;Gnd Spd alarm	Level 2	17	11	12	27	67
Eng Spd alarm	Level 2	61	64	27	40	192
Gear Select alarm	Level 2	4	6		3	13
Ground Spd alarm	Level 2				2	2
Lt F Brk Temp alarm	Level 2		1		2	3
Lt R Brk Temp alarm	Level 2	3	70	72	10	155
ParkBk alarm	Level 2	18	47	110	50	225
Rt F Brk Temp alarm	Level 2	46	5	5	62	118
Rt R Brk Temp alarm	Level 2	4	2	3	8	17
Rt R Brk Temp alarm	Level 3				2	2
Total		197	284	298	235	1014

7. Make the reports visible to those responsible for and affected by the results (Training, Operations, and Maintenance departments). Graphs can be placed in hallways, lunch room, pre-shift meeting room, etc. to communicate results and performance by shift / crew. Doing so can be a key contributor in keeping overall event numbers low, as site personnel

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is aware of the current situation and challenges, and can work together to resolve potential issues.

8. Determine whether any of the issues identified in the data are operator-specific or broader across multiple operators and/or crews. If the former is true, awareness, understanding, and instruction should be given to the operator to correct the situation. If the latter is true, training should be coordinated with the training instructor and/or Cat dealer to provide the necessary instruction on proper operating technique for the operating staff.

4.0 Benefits

Although the benefits will vary by mine site, the following may be expected:

- Reduction in number of VIMS[™] events per machine
- Improved physical availability
- Improved MTBS (Mean Time Between Stoppages)
- Tons per hour increase

One customer site that anonymously contributed to this Best Practice has seen a large reduction in the overall number of events since its inception of in-depth VIMS[™] analysis by the Training department about five years ago. Also, interestingly enough, at one time the site stopped preparing these reports. However, due to the health numbers beginning to rise, the reports were prepared again to bring the number of events back down. In other words, the site has definitely seen tangible value in utilizing VIMS[™] information as part of its Training program.

5.0 Resources Required

- Laptop computer(s) loaded with VIMSpc or VIMS 2011A software program for those individuals who will be analyzing the data.
- If performing manual downloads, the VIMS[™] cable will be needed: 127-9797 (used for machines with VIMS[™] 68k and VIMS[™] ABL), or 305-5528 (used for machines with VIMS[™] 3G).
- Equipment training analyst with application and/or operating experience, as well as familiarity with VIMS[™].

In addition to these items, team work is also a key ingredient required to make this initiative work. It takes the Operations Trainer, Operations Supervisors, Equipment Operators, and the Maintenance department working together to resolve issues that are encountered.

6.0 Supporting Attachments / References

VIMS[™] information – <https://mining.cat.com/vims>

7.0 Related Best Practices

None.

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8.0 Acknowledgements

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Cashman Equipment is specially acknowledged for contributing information related to this best practice. Cat Global Mining extends its appreciation to Cashman Equipment for sharing this idea with the extended Cat mining team.

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